

Chapter: 1 Introduction of statistics and Presentation of Data

(2022-Fall), Pre-Board, NAST

prabin ch, NAST

1. An article in environment concentration and Toxicology report the amount of cadmium in sea scallops observed at a number of different station in North Atlantic water. The observed values are as follows,

5.1, 14.1, 14.7, 10.8, 6.5, 7.7, 14.1, 9.5, 3.7, 8.9,
7.9, 7.9, 4.5, 10.1, 5.0, 9.6, 5.5, 5.1, 11.4, 8.0,
12.1, 7.5, 8.5, 13.1, 6.4, 18.0, 27.0, 18.9, 10.8, 13.1,
8.4, 16.9, 2.7, 9.6, 4.5, 12.4, 5.5, 12.7, 17.1, 20.5,
22.3

i) Construct a frequency distribution for the dataset using 0.0 - 4.0 and 4.0 - 8.0 as the first and second class interval respectively.

ii) Draw the histogram and frequency curve of the above frequency distribution.

Sol Here,

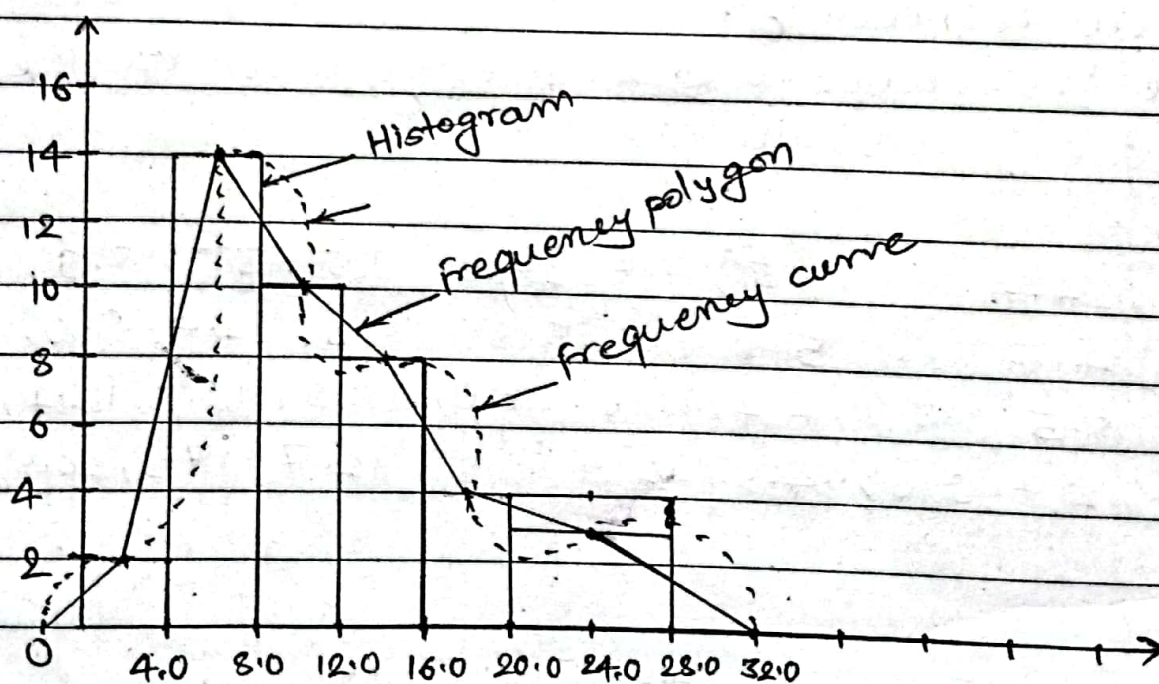
5.1, 14.1, 14.7, 10.8, 6.5, 7.7, 14.1, 9.5, 3.7, 8.9,
7.9, 7.9, 4.5, 10.1, 5.0, 9.6, 5.5, 5.1, 11.4, 8.0,
12.1, 7.5, 8.5, 13.1, 6.4, 18.0, 27.0, 18.9, 10.8, 13.1,
8.4, 16.9, 2.7, 9.6, 4.5, 12.4, 5.5, 12.7, 17.1, 20.5,
22.3

$$\text{No. of class} = 1 + 3.322 \log(41) = 6.3577 \\ \approx 6$$

i) Frequency distribution for the data:

Amount of cadmium	Tally marks	frequency
0.0 - 4.0		2
4.0 - 8.0		14
8.0 - 12.0		10
12.0 - 16.0		8
16.0 - 20.0		4
20.0 - 28.0		3
	$N = 41$	

ii) Histogram and frequency curve:



2012-Spring

Q. Given below the pre-tax monthly income (in rupees) of the workers of FEWA hydropower in 2000.

Income more than ('000' Rs.)	7	6	5	4	3	2	1	0
No. of workers	2	8	10	15	25	40	55	60

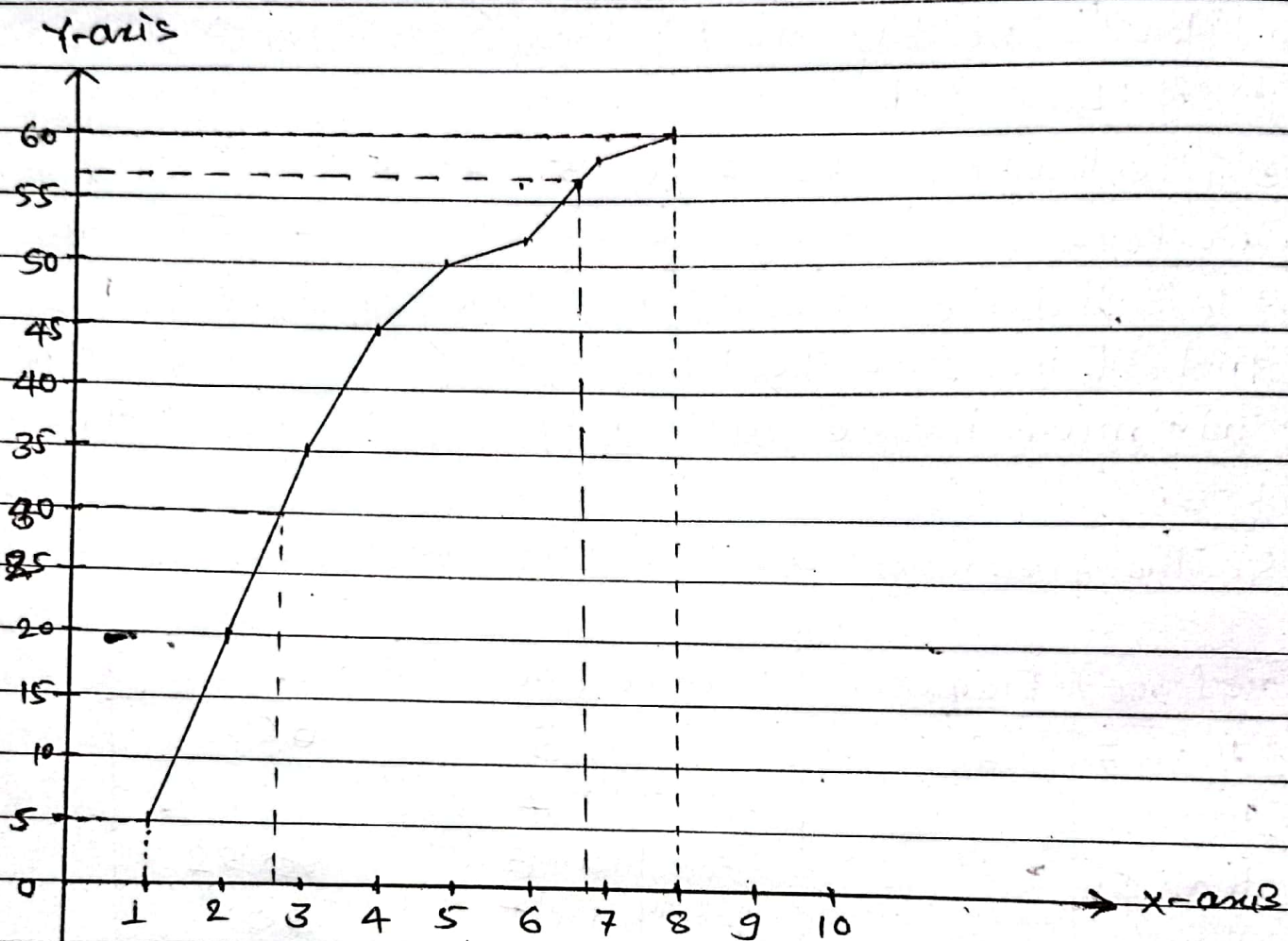
Draw less than ogive curve and hence find out:

- The highest income of the lowest 50% of the worker.
- The minimum income earned by the top 5% of the workers.

~~sol~~ Here, The frequency distribution can be written as:

Income ('000')	Frequency	Income ('000')	c.f.
more than 7	2	less than 8	60
more than 6	8	less than 7	58
more than 5	10	less than 6	58 52
more than 4	15	less than 5	50
more than 3	25	less than 4	45
more than 2	40	less than 3	35
more than 1	55	less than 2	20
more than 0	60	less than 1	5

For less than ogive curve, plotting the points (8, 60), (7, 58), (6, 52), (5, 50), (4, 45), (3, 35), (2, 20) & (1, 5).



i) Total no. of workers, $N = 60$

$$50\% \text{ of workers} = \frac{50}{100} \times 60 = 30$$

The highest income of the lowest 50% of the workers

$$= 2 + \frac{2}{3} \times (3-1) \times 1000 = \text{Rs. } 2670$$

ii) The minimum income earned by the top 5% of the worker:

$$= \frac{5 \times 60}{100} = \text{Rs. } 3 \text{ (i.e. } 60 - 57 = 3)$$

Exam Asked Questions

① The marks obtained by 50 students in numerical methods are given as

65	36	49	84	79	56	28	43	67	86
43	78	37	40	68	72	70	55	62	82
88	50	60	56	57	46	39	57	22	65
59	48	76	74	80	69	51	40	56	45
35	21	62	52	63	32	86	64	53	34

i) Construct a frequency distribution from the above. What additional information did you receive from frequency distribution that you didn't receive from the original raw data?

ii) Represent the data by means of cumulative frequency curve. Identify the median and first quartile from the curve. Also interpret the result of first quartile.

Sol We have,

$$N = 50$$

$$S = 21$$

$$L = 88$$

$$\text{No. of class} = 1 + 3.322 \log(50) = 6.64 \approx 7$$

i) Frequency distribution Table:

Marks	Tally marks	Frequency
20-30	III	3
30-40	III III	9
40-50	III II	7
50-60	III III I	11
60-70	III III	10
70-80	III I	6
80-90	IIII	4
	$N = 50$	

From the frequency distribution table, we see that the no. of students getting 40 to 70 is more higher than getting less than 40 and more than 70.

i.e. The no. of students getting average marks is more higher than above and below the average.

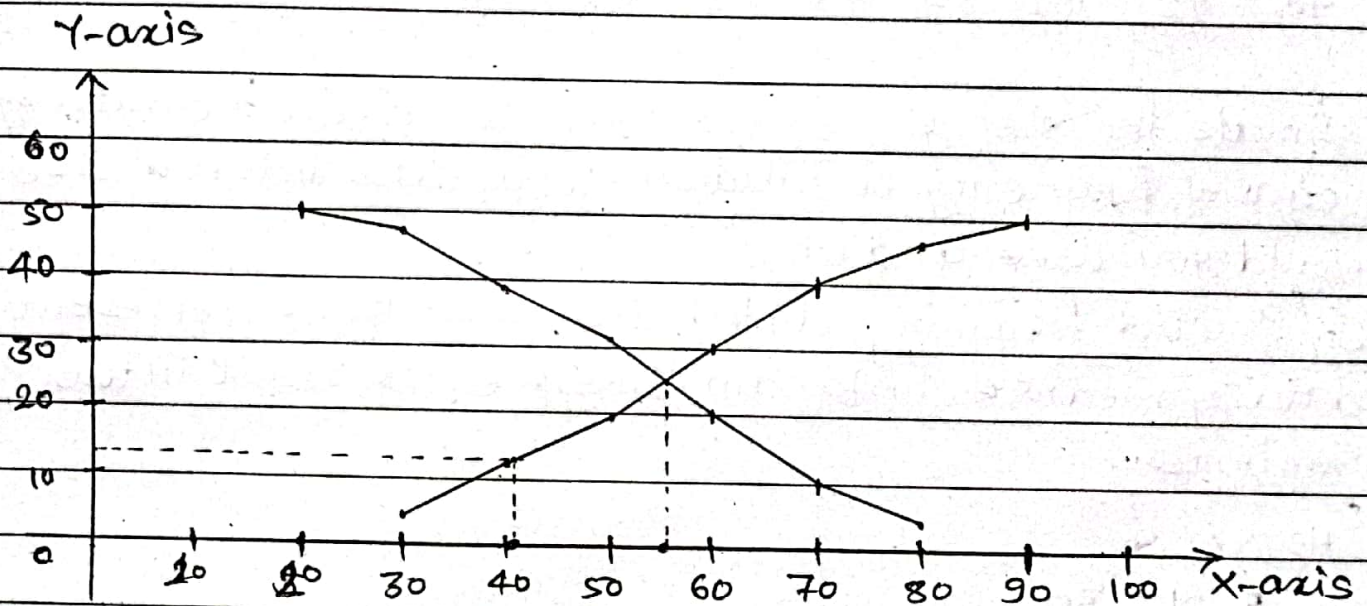
ii) Here,

Marks	Frequency	Marks	C.f.	Marks	C.f.
20-30	3	more than 20	50	less than 30	4
30-40	9	more than 30	47	less than 40	12
40-50	7	more than 40	40	less than 50	19
50-60	11	more than 50	29	less than 60	30
60-70	10	more than 60	19	less than 70	40
70-80	6	more than 70	10	less than 80	46
80-90	4	more than 80	4	less than 90	50
Total	$N = 50$				

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For more than ogive, plotting the points $(20, 50)$, $(30, 47)$, $(40, 38)$, $(50, 31)$, $(60, 20)$, $(70, 10)$, $(80, 4)$ in the graph.

For less than ogive, plotting the points $(30, 4)$, $(40, 12)$, $(50, 19)$, $(60, 30)$, $(70, 40)$, $(80, 46)$, $(90, 50)$ in the graph.



From the above graph, median = 56 Ans

For first quartile, $= \frac{N}{4} = \frac{50}{4} = 12.5$

From graph, first quartile = ~~40~~ 42

Thus, the minimum marks obtained by first 25% students is 42 Ans

2020 Fall

① The high temperature $^{\circ}(\text{F})$ in a random sample for 30 days readings during summer season in Nepal Terai recorded as below.

94 78 88 95 102 89 88 88 99 105

78 97 88 82 98 87 94 96 86 87

94 92 102 84 75 69 88 94 84 89

i) Estimate the step you would take to convert the data by means of frequency distribution. What class interval size would you use and why?

ii) Construct a frequency distribution and hence represent the data by means of histogram. Interpret the result in two sentences.

~~So~~ Here,

$$N = 30$$

$$S = 69$$

$$L = 105$$

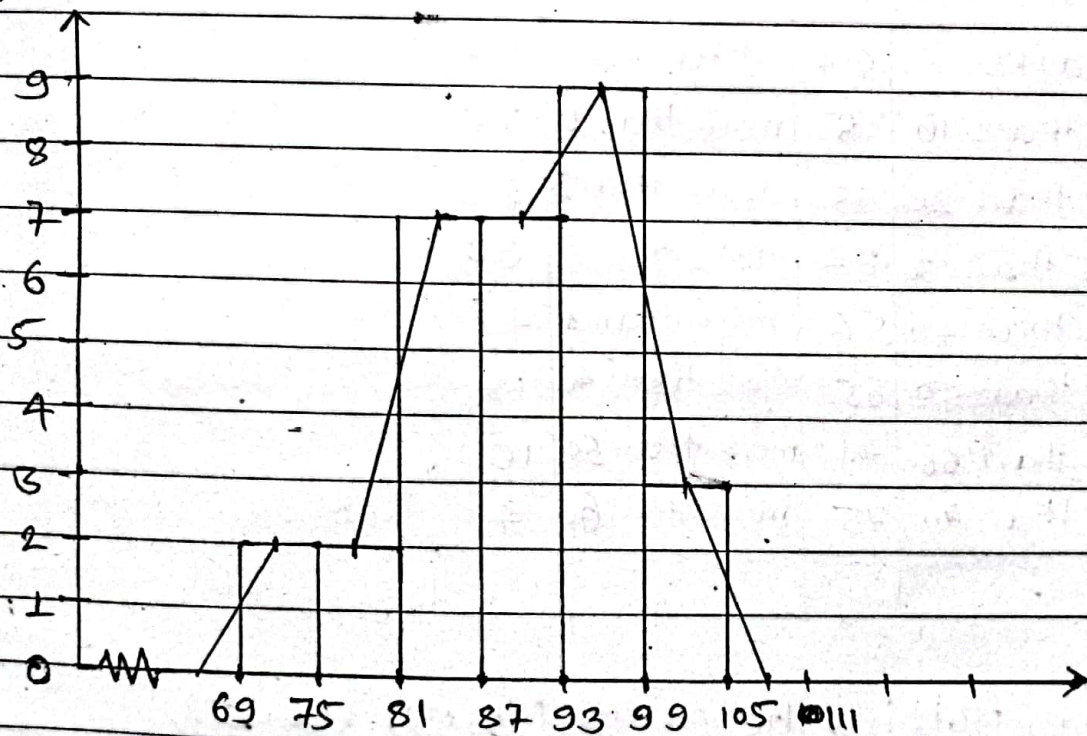
$$\therefore \text{No. of class} = 1 + 3.322 \log_{10}(30) = 5.9069 \approx 6$$

$$\therefore \text{Size of class interval} = \frac{L - S}{\text{No. of class}} = \frac{105 - 69}{6} = 6$$

i) Therefore, the total no. of class will be 6 and class interval size would be 6 and the frequency distribution is shown below:

ii) Frequency distribution and histogram:

Temperature	Tally marks	Frequency
69-75		2
75-81		2
81-87		7
87-93		7
93-99		9
99-105		3
	$N = 30$	



2019-spring

④ From the following frequency distribution,

Obtained marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70
No. of students	5	10	17	24	9	6	4

Construct an ogive curve that will help you the answer to find the no. of students securing marks:

i) less than 35 marks

ii) Between 20 and 50 marks

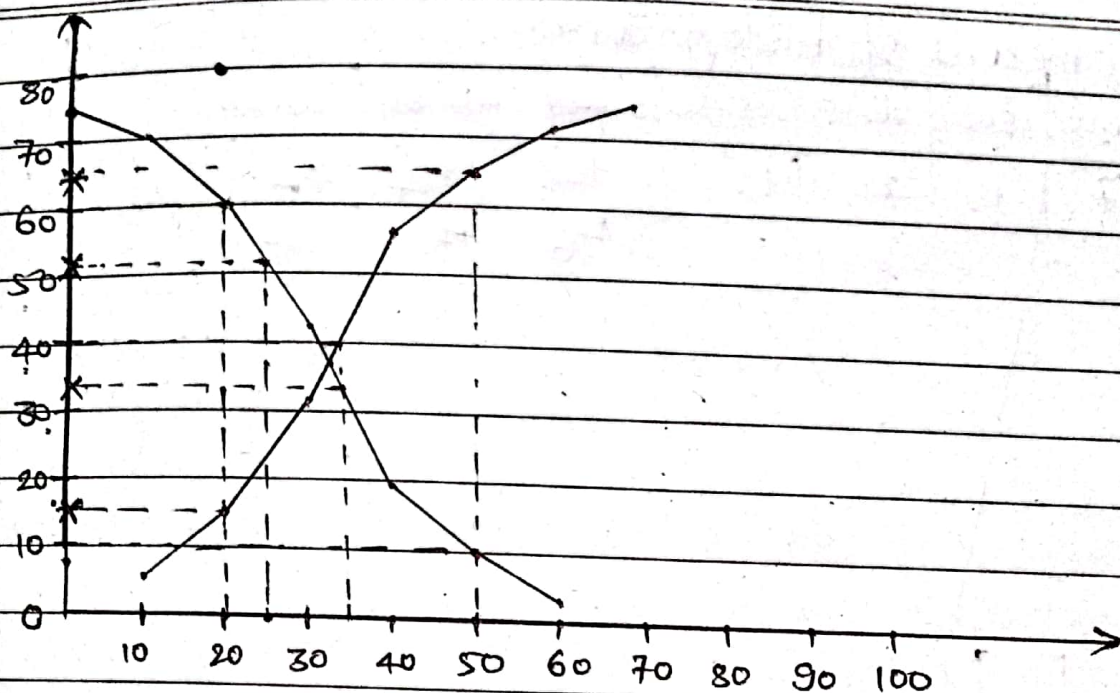
iii) More than 25 marks.

sol Here,

Marks	f	Marks	c.f	Marks	c.f.
0-10	5	less than 10	5	more than 0	75
10-20	10	less than 20	15	more than 10	70
20-30	17	less than 30	32	more than 20	60
30-40	24	less than 40	56	more than 30	43
40-50	9	less than 50	65	more than 40	19
50-60	6	less than 60	71	more than 50	10
60-70	4	less than 70	75	more than 60	4
	N=75				

For less than ogive, plotting the points, (10, 5), (20, 15), (30, 32), (40, 56), (50, 65), (60, 71), (70, 75) in the graph.

i) For more than ogive, plotting the points (0, 75), (10, 70), (20, 60), (30, 43), (40, 19), (50, 10), (60, 4) in the graph.



i) The no. of students securing marks less than 35 is ~~34~~ 31.

ii) The no. of students securing marks between 20 and 50 is: ~~75 - 60 = 15~~, $65 - 15 = 50$

iii) The no. of students securing marks more than ²⁵ ~~25~~ 31, 52. ~~Ans~~

2018-Spring

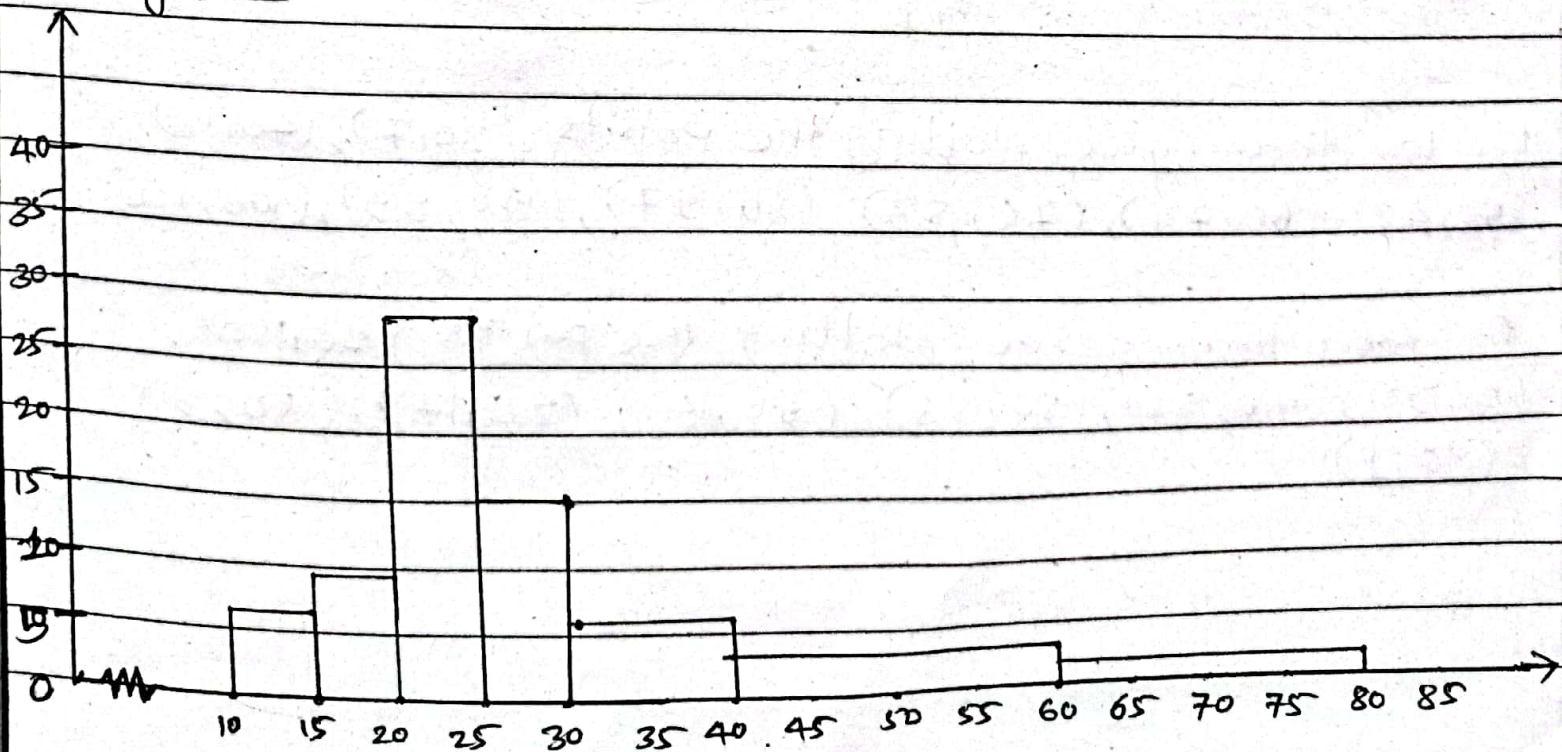
7. Represent by means of histogram.

wages:	10-15	15-20	20-25	25-30	30-40	40-60	60-80
No. of workers	7	9	28	15	12	12	8

Here,

wages	No. of workers;
10-15	7
15-20	9
20-25	28
25-30	15
30-40	$12 \div 2 = 6$
40-60	$12 \div 4 = 3$
60-80	$8 \div 4 = 2$
	$N = 91$

Histogram:



18) Draw less than and more than ogive for the following frequency distribution:

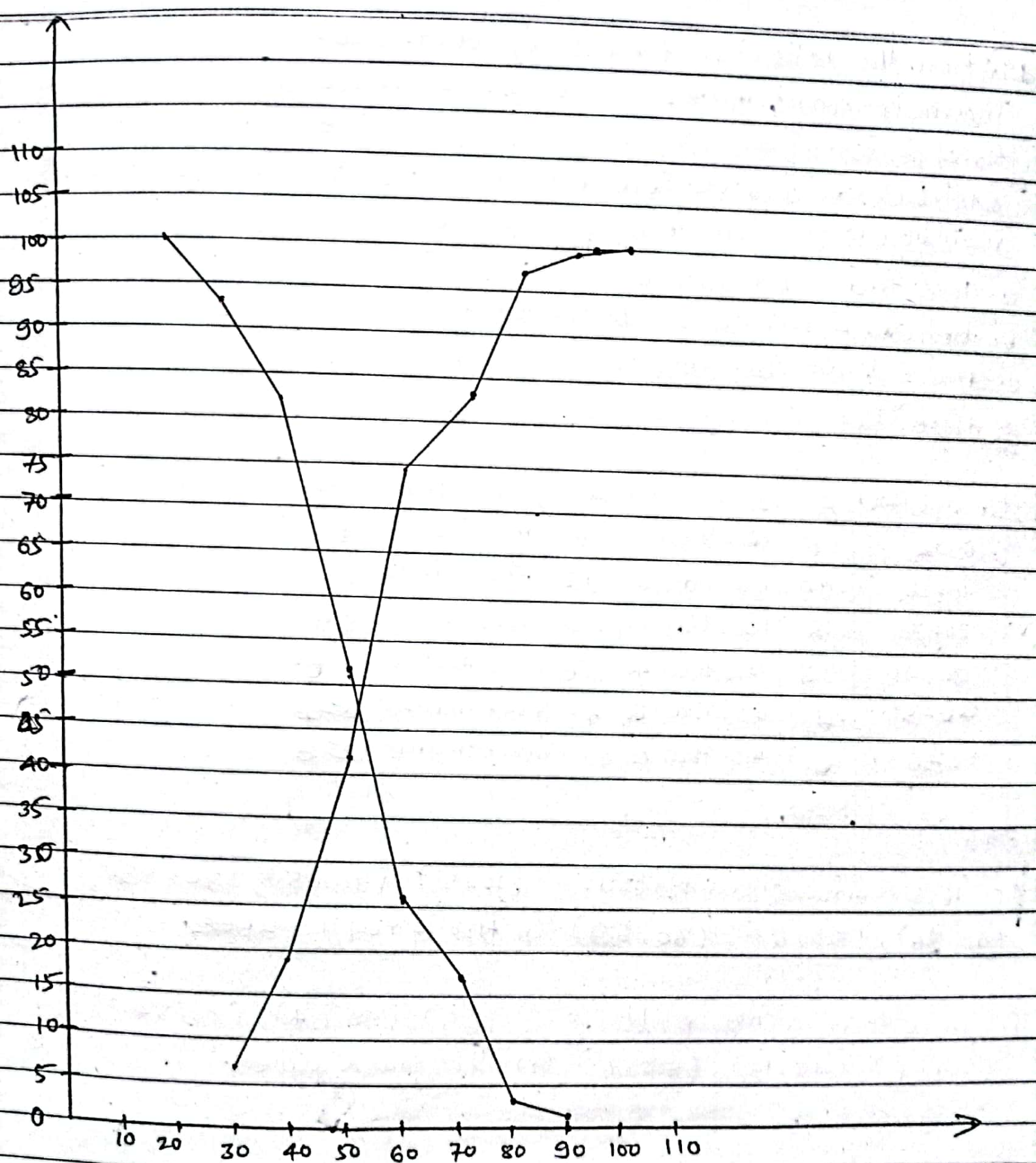
Marks	20-30	30-40	40-50	50-60	60-70	70-80	80-90	9-100	
No. of students	7	11	24	32	9	14	2	1	

Here,

Marks	Frequency	Marks	c.f.	Mark	c.f.
20-30	7	less than 30	7	more than 20	100
30-40	11	less than 40	18	more than 30	93
40-50	24	less than 50	42	more than 40	82
50-60	32	less than 60	74	more than 50	58
60-70	9	less than 70	83	more than 60	26
70-80	14	less than 80	97	more than 70	17
80-90	2	less than 90	99	more than 80	3
90-100	1	less than 100	100	more than 90	1
	N =				

For less than ogive, plotting the points (30, 7), (40, 18), (50, 42), (60, 74), (70, 83), (80, 97), (90, 99), (100, 100)

For more than ogive, plotting the points (20, 100), (30, 93), (40, 82), (50, 58), (60, 26), (70, 17), (80, 3), (90, 1).



(19) From the following frequency distribution

Income (Rs. 000)	0-10	10-20	20-30	30-40	40-50	50-60
No. of person	5	10	18	23	7	6

Construct an ogive that will help you the answer find the number of persons having income:

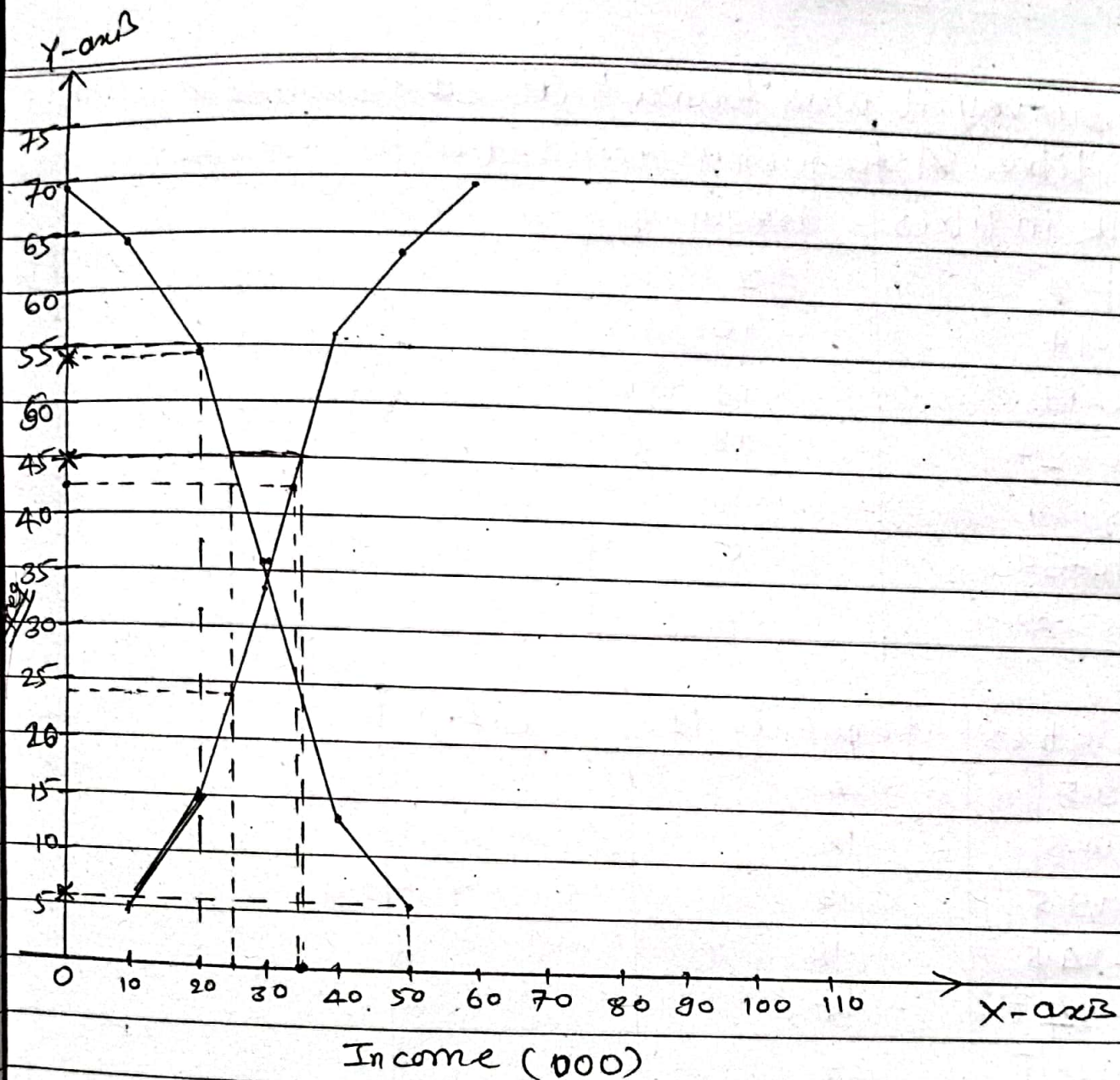
- less than Rs. 35000
- between Rs. 20,000 and Rs. 50,000
- more than Rs. 25,000.

So Here,

Income (000)	f	Income (000)	cf	Income (000)	c.f.
0-10	5	less than 10	5	more than 0	69
10-20	10	less than 20	15	more than 10	64
20-30	18	less than 30	33	more than 20	54
30-40	23	less than 40	56	more than 30	36
40-50	7	less than 50	63	more than 40	13
50-60	6	less than 60	69	more than 50	6
	N=69				

For less than ogive, plotting (10, 15), (20, 33), (30, 54), (40, 63), (50, 69) in the graph paper

For more than ogive, plotting (0, 69), (10, 64), (20, 54), (30, 36), (40, 13), (50, 6) in the graph paper



i) The number of persons having income less than Rs. 35000 is 45. Ans

ii) The number of persons having income between Rs. 20,000 and Rs. 30,000 is: $55 - 25 = 30$ Ans

iii) The number of ~~student~~ persons having income more than Rs. 25,000 is 45. Ans

2023-Spring

1.a) In a survey, it was found that 64 families bought milk in litre. A frequency distribution is given as:

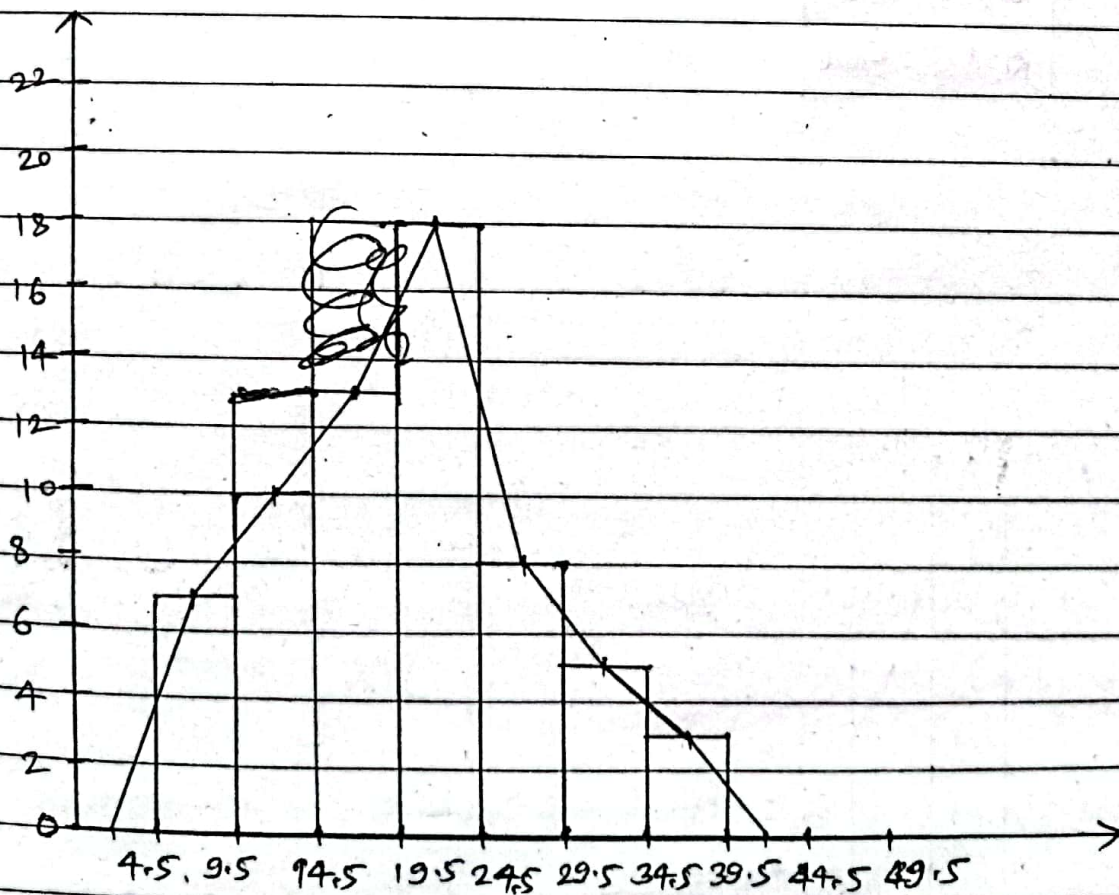
Milk in litres	Frequency
5-9	7
10-14	10
15-19	13
20-24	18
25-29	8
30-34	5
35-39	3

89 Here,

Milk in litres	frequency (f)	c.f.
4.5 - 9.5	7	
9.5 - 14.5	10	
14.5 - 19.5	13	
19.5 - 24.5	18	
24.5 - 29.5	5	
29.5 - 34.5	3	
34.5 - 39.5		
Total:		

Milk in litres	frequency (f)	c.f.
4.5-9.5	7	7
9.5-14.5	10	17
14.5-19.5	13	30
19.5-24.5	18	48
24.5-29.5	8	56
29.5-34.5	5	61
34.5-39.5	3	64
	$N = 64$	

Histogram and frequency polygon.



2022-Fall

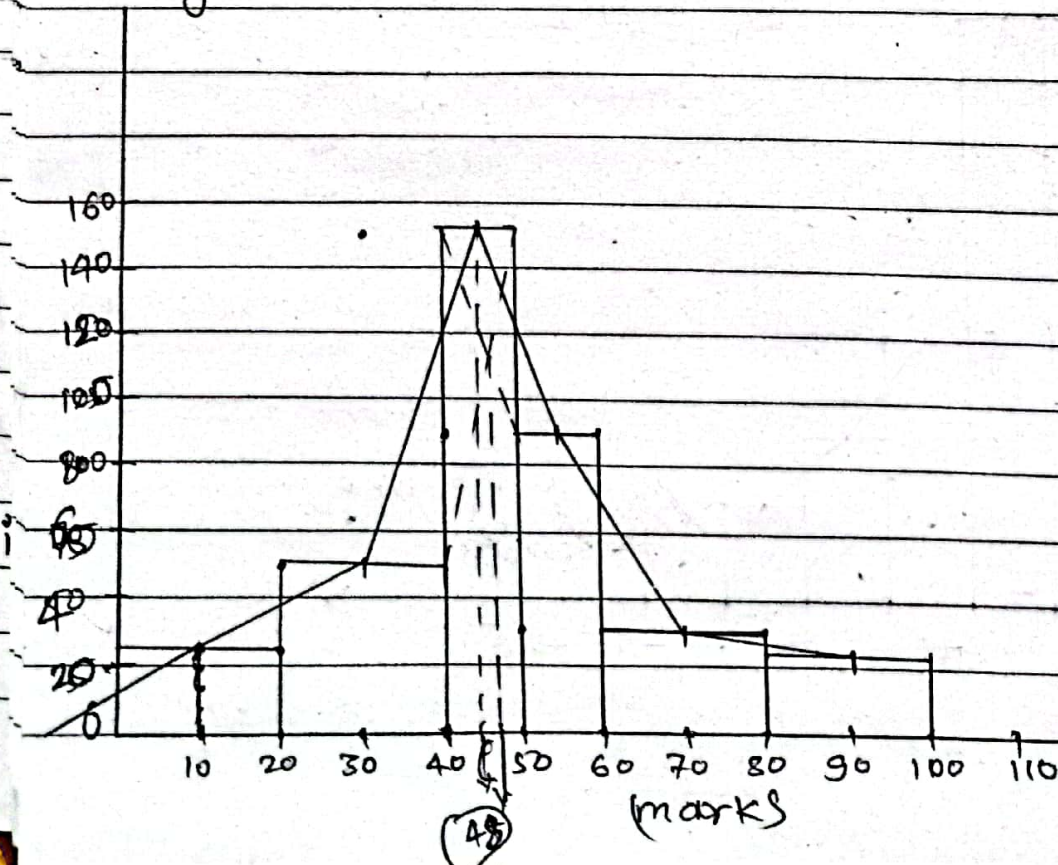
1.a) From the following distribution of 500 students of a college, represent the data by histogram and locate the mode.

Marks	0-20	20-40	40-50	50-60	60-80	80-100
No. of students	50	100	150	90	60	50

~~809~~

Marks	frequency
0-20	$50 \div 2 = 25$
20-40	$100 \div 2 = 50$
40-50	150
50-60	90
60-80	$60 \div 2 = 30$
80-100	$50 \div 2 = 25$
	N=500 500

* Histogram,

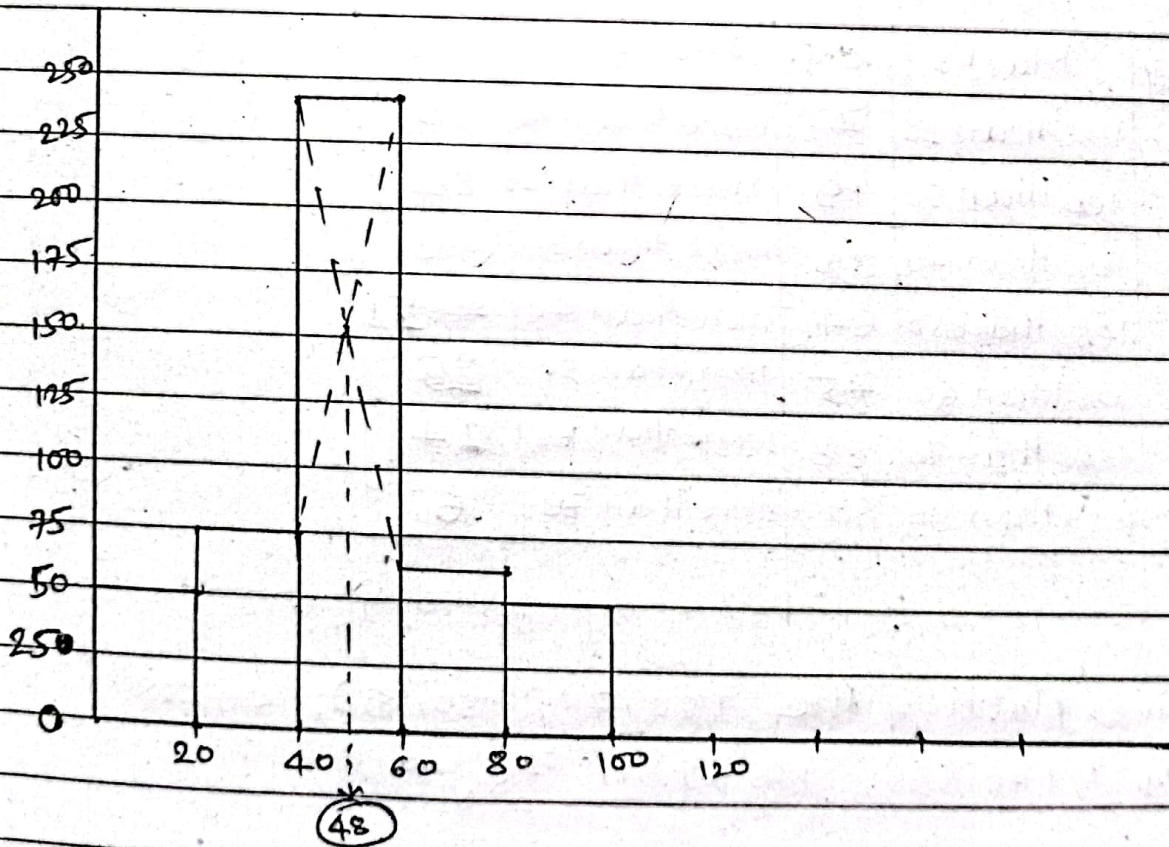


$$\begin{aligned} \cos^2 v &= \frac{1 + \cos 2v}{2} \\ \cos^3 v &= \frac{3 \cos v + \cos 3v}{4} \\ \sin^2 v &= \frac{1 - \cos 2v}{2} \\ \sin^3 v &= \frac{3 \sin v - \sin 3v}{4} \end{aligned}$$

$$\begin{aligned} dx &= \cos v \, dv \\ dx &= -\sin v \, dv \end{aligned}$$

or

Marks	frequency
0-20	50
20-40	100
40-60	240
60-80	60
80-100	250
	$N = 300$



Hence the mode of the ~~data~~ is 48

2021-Fall

1. a) From the following frequency distribution;

Marks obtained	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of students	7	12	19	26	11	8	6

Construct an ogive that will help you the answers to find the number of students securing marks.

i) Less than 45 marks.

ii) Between a 30 and 70 marks.

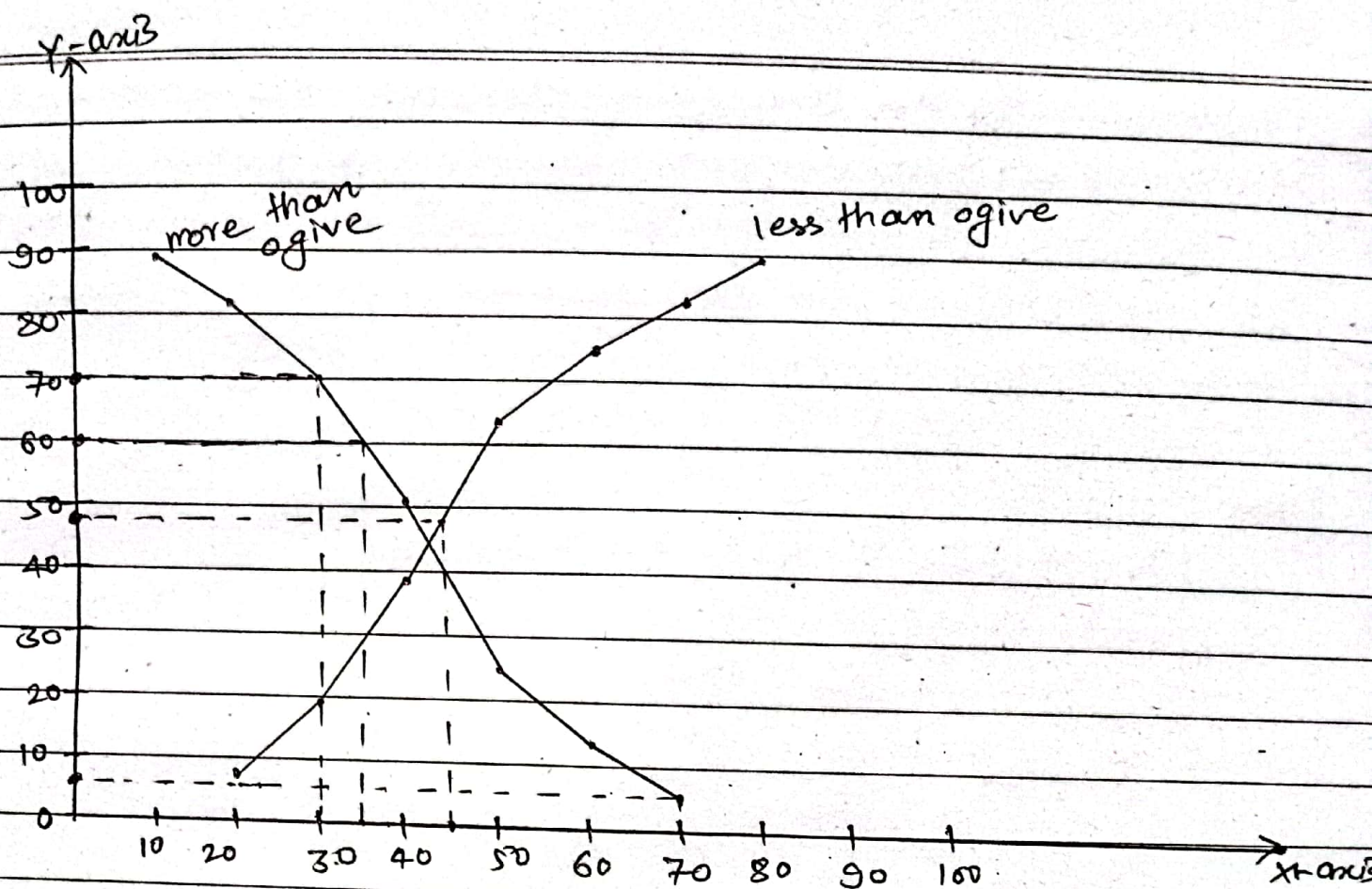
iii) More than 35 marks.

Here,

Marks	frequency	Marks	c.f.	Marks	c.f.
10-20	7	less than 20	7	more than 10	89
20-30	12	less than 30	19	more than 20	82
30-40	19	less than 40	38	more than 30	70
40-50	26	less than 50	64	more than 40	41 51
50-60	11	less than 60	75	more than 50	25
60-70	8	less than 70	83	more than 60	14
70-80	6	less than 80	89	more than 70	6
	N =				

For less than ogive, plotting the $(20, 7)$, $(30, 19)$, $(40, 38)$, $(50, 64)$, $(60, 75)$, $(70, 83)$, $(80, 89)$ in the graph.

For more than ogive, plotting the $(10, 89)$, $(20, 82)$, $(30, 70)$, $(40, 51)$, $(50, 25)$, $(60, 14)$, $(70, 6)$ in the graph.



- i) The number of students securing marks less than 45 marks is 48. Ans
- ii) The number of students securing marks between 30 & 70 marks is $70 - 60 = 10$. Ans
- iii) The number of students securing marks more than 35 marks is 60 . Ans